Chapter 2: Describing Distributions with Numbers

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Part I: Exploring Data (The Basics)

The Limits of Human Perception

- In Chapter 1 we learned the first rule of modern data analysis: **always plot your data**.
- Visualizations (histograms, dotplots) allow our eyes to quickly spot shapes, trends, bimodal clusters and data entry errors.
- **The Problem:** Human eyes are subjective!
  - If you ask two scientists to point to the "center" of a skewed histogram, they will point to two different spots.
- **The Solution:** We need to translate our visual descriptions - Shape, Outliers, Center and Spread (SOCS) - into objective, reproducible statistical summaries.

Measuring Center: What is "Typical"?

Defining the Center

When a biologist asks *"What is the typical length of a pine needle?"* or *"What is the normal number of flood events on the coast?"*, they are asking for a measure of **center**.

Statistics gives us two primary ways to define "typical" and they behave very differently depending on the shape of the data:

- The **Mean** ($\bar{x}$)
- The **Median** (M)

The Mean ($\bar{x}$)

The most commonly reported measure of center is the ordinary arithmetic average or **mean**. To find it, add all values together and divide by the number of observations (n).

If the n observations are $x_1, x_2, \ldots, x_n$, their sample mean is

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

We use the uppercase Greek letter sigma ($\sum$) to mean "add them all up". So the compact formula is

$$\bar{x} = \frac{1}{n} \sum x_i$$

Notation Note: Sample vs. Population

- The subscripts (x_i) just keep the n observations distinct; they do not necessarily indicate chronological order.
- The bar over the x indicates the mean of all the x-values.

Important Distinction

We use $\bar{x}$ to represent the mean of a *sample* of data. Later in the course, we will use the Greek letter μ (mu) to represent the theoretical mean of an entire *population*. Whenever you see $\bar{x}$ in a paper, the authors are discussing sample data.

Example 2.3: Mean of Pine Needle Lengths

Let's find the mean of the needle length for a sample of $n = 15$ Aleppo pine needles. The sample lengths are given in the table below.

Obs (i)	x_i **(cm)**	**Obs (i)**	x_i **(cm)**	**Obs (i)**	x_i **(cm)**
1	7.2	6	9.0	11	10.2
2	7.6	7	9.0	12	10.9
3	8.5	8	9.3	13	11.3
4	8.5	9	9.4	14	12.1
5	8.7	10	9.4	15	12.8

Example 2.3: Calculating the Mean

Step 1: Sum the values ($\sum$)

$$\sum_{i=1}^{15} x_i = 7.2 + 7.6 + \cdots + 12.8 = 143.9 \text{ cm}$$

Step 2: Divide by n

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{15} x_i = \frac{143.9}{15} \approx 9.59 \text{ cm}$$

Takeaway: The he formula reminds us of a critical fact that the mean mathematically incorporates the actual numerical value of every single observation.

Example 2.3: Calculating the Mean in R

We can use R to find the mean instantly.

R Code

```
# Create a vector of the 15 pine needle lengths
pine_needles <- c(7.2, 7.6, 8.5, 8.5, 8.7, 9.0, 9.0, 9.3,
                  9.4, 9.4, 10.2, 10.9, 11.3, 12.1, 12.8)

# Calculate the mean
mean(pine_needles)
```

Output: `[1] 9.593333`

The Median (M)

- The **median** is the exact geometric midpoint of a distribution.
- It is the 50th percentile. This is the number such that exactly half (50%) of the observations are smaller and the other half are larger.

How to locate the median (M)

1. **Sort** all observations in order from smallest to largest.
2. If n is **odd**, the median is located at position $\frac{n+1}{2}$ of the ordered data.
3. If n is **even**, this lands between two numbers, and you take their average.

Software Note: The Median in R

Calculating the Median

In R, the median is calculated using the simple `median()` function.

Unlike calculating by hand where we must sort the data first, R handles the sorting, locating and averaging (if n is even) automatically behind the scenes.

Example 2.1: Finding the Median (Odd n)

What is the median length for our 15 Aleppo pine needles?

1. **Sort the data:**
   21.2, 21.6, 21.7, 23.1, 23.7, 24.2, 24.2, 25.5, 26.6, 26.8, 28.9, 29.0, 29.7, 29.7, 30.2, 32.5, 33.7, 33.7
2. **Locate the median:**

$$\text{Location} = \frac{n+1}{2} = \frac{15+1}{2} = 8$$

3. **Find the middle:**
   The 8th observation in the sorted list is 9.3 cm.

Result: Median $(M) = 9.3$ cm.

Example 2.1: Finding the Median in R (Odd n)

R Code

```
# Using our pine_needles vector from earlier
median(pine_needles)
```

Output: `[1] 9.3`

Example 2.2: Finding the Median (Even n)

Now we have the lengths of 18 needles from the Torrey pine species.

1. **Sort the data:**
   21.2, 21.6, 21.7, 23.1, 23.7, 24.2, 24.2, 25.5, **26.6, 26.8**, 28.9, 29.0, . . . (18 total)
2. **Locate the median:**

$$\text{Location} = \frac{n+1}{2} = \frac{18+1}{2} = 9.5$$

3. **Find the middle:**
   Location 9.5 means exactly halfway between the 9th and 10th observations: $\frac{26.6+26.8}{2}$.

Result: Median $(M) = 26.7$ cm.

Example 2.2: Finding the Median in R (Even n)

R Code

```
# Create a vector of the 18 Torrey pine needle lengths
torrey_needles <- c(21.2, 21.6, 21.7, 23.1, 23.7, 24.2,
                    24.2, 25.5, 26.6, 26.8, 28.9, 29.0,
                    29.7, 29.7, 30.2, 32.5, 33.7, 33.7)

# Calculate the median
median(torrey_needles)
```

Output: `[1] 26.7`

Comparing Mean and Median: Resistance

If both the mean and the median measure the "center," why do we need both? **Because they react differently to extreme biology.**

- A statistical measure is called **resistant** (or robust) if it is not strongly influenced by a few extreme observations (outliers).
- **The Median is Resistant:** It is influenced *only* by the total number of data points and the physical order of the numbers.
- **The Mean is Non-Resistant:** Because the mean adds up the numerical value of every single observation, it cannot resist the pull of extreme outliers.

Example: The "Long-Hauler" Patient

Patients tracking days for Heart Rate Variability (HRV) to return to baseline.

Normal Sample

Sorted: 4, 6, 7, 8, 9

- **Median (M):** 7 days
- **Mean ($\bar{x}$):** 6.8 days

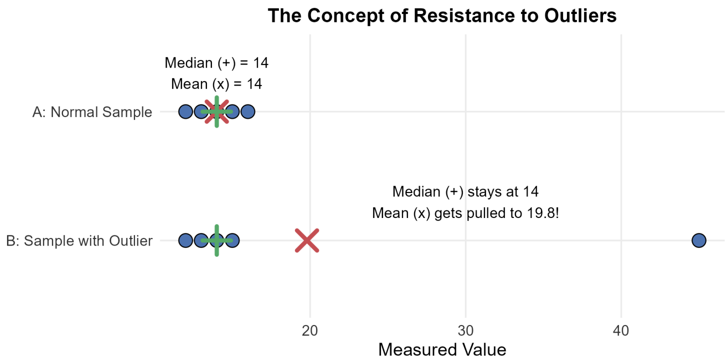
(For symmetric data, they are very close together).

Sample with Outlier

Sorted: 4, 6, 7, 8, **45**

- **Median (M):** 7 days *(Did not change!)*
- **Mean ($\bar{x}$):** 14.0 days *(Drastically pulled right!)*

Visualizing Resistance to Outliers



- **The Median is Resistant:** Changing the largest value does not change the physical midpoint.
- **The Mean is Non-Resistant:** The massive outlier acts like a heavy weight on the scale, dragging the mean toward it.

Shape Determines the Center

Because of resistance, the shape of your distribution dictates how the mean and median will relate to each other:

Shape	Mathematical Relationship
Symmetric	Mean $\approx$ Median (Both sit in the center)
Skewed Right	Mean $>$ Median (Extreme high values pull mean to the right)
Skewed Left	Mean $<$ Median (Extreme low values drag mean to the left)

Example 2.4: Flooding in Coastal Communities

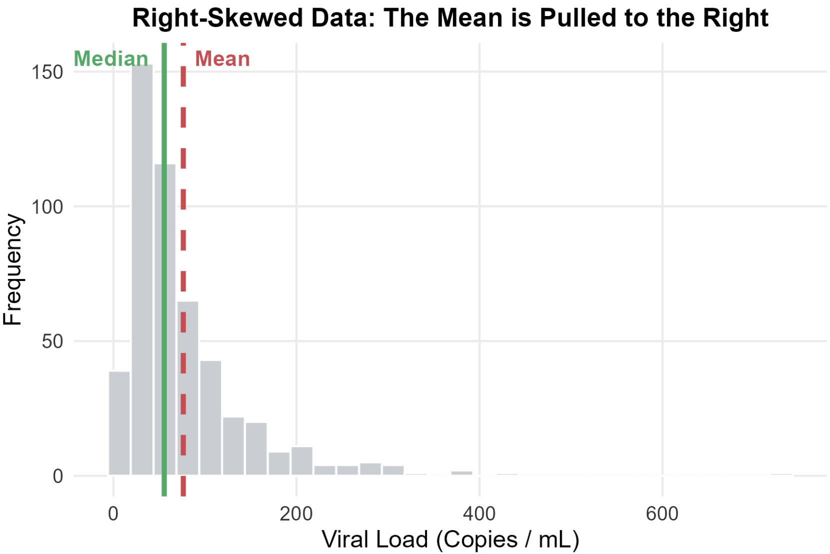
Number of flood events recorded (2009–2013) in 52 Atlantic coastal communities.

- **Shape:** Noticeably skewed to the right (most communities have moderate floods, a few hotspots have massive numbers).
- **Median (*M*):** 37.5 flood events
- **Mean ($\bar{x}$):** 55.3 flood events

If the hardest-hit community had a typo, logging **2,500** floods instead of 250:

- The median would remain exactly **37.5**.
- The mean would skyrocket to **98.6**.

Visualizing Right-Skewed Data



Biological Application: Survival Times

- Many biological variables (like viral loads or survival times) are **right-skewed**.
- Most patients survive a typical duration, but a few "super-responders" survive for years, creating a long right tail.
- **Doctors** generally report the *median* to patients because it represents the typical midpoint experience.
- **Hospitals** must calculate the *mean* to project the total staffing and financial resources needed to cover all patient days.
- *Takeaway:* Both measures measure "center" in different ways, and both are scientifically useful!

Measuring Spread: Why Center Isn't Enough

- Saying that the median penguin weighs **4.5 kg** tells us the center.
- But it doesn't tell us how widely penguin weights vary!
- Are most penguins exactly 4.5 kg, or do they range wildly from 2 kg to 8 kg?
- **Rule:** A measure of center must *always* be accompanied by a measure of **spread** (variability).

The Quartiles (Q_1 and Q_3)

Just as the median divides the data exactly in half, the quartiles divide the data into exactly four equal quarters.

- **Quartile 1 (Q_1):** The median of the lower half of the data. Exactly **25%** of the data falls below Q_1.
- **Quartile 3 (Q_3):** The median of the upper half of the data. Exactly **75%** of the data falls below Q_3 (and 25% is above it).

How to Calculate the Quartiles by Hand

1. Sort the data and find the overall Median (M).
2. Locate the lower half of the data. *(Note: If n is odd, do not include the overall median in this lower half).* The median of this lower half is Q_1.
3. Locate the upper half of the data. The median of this upper half is Q_3.

Data Entry Tip

Be careful when several observations take the exact same numerical value. Include all the duplicate observations in your list and apply the rules just as if they all had distinct values!

Example 2.5: Finding Quartiles (Odd n)

Aleppo pine needles ($n = 15$), ordered:

Lower Half	**Median**	**Upper Half**
7.2, 7.6, 8.5, **8.5**, 8.7, 9.0, 9.0	**9.3**	9.4, 9.4, 10.2, **10.9**, 11.3, 12.1, 12.8

- Overall Median (M) = **9.3 cm**.
- Q_1**:** The median of the 7 observations to the left. The middle of 7 is the 4th value $\rightarrow$ **8.5 cm**.
- Q_3**:** The median of the 7 observations to the right. The middle of 7 is the 4th value $\rightarrow$ **10.9 cm**.

Notice: Q_3 would still be 10.9 cm even if the maximum needle was 50 cm! Quartiles are highly resistant.

Example 2.5: Finding Quartiles in R

R provides several ways to find quartiles. The most common is the `summary()` function, which gives the quartiles, the extremes, and the mean all at once.

R Code

```
summary(pine_needles)
```

Output:

```
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 7.200   8.600   9.300   9.593  10.550  12.800
```

Example 2.6: Finding Quartiles (Even n)

Torrey pine needles ($n = 18$), ordered:

Lower Half	**Median Area**	**Upper Half**
21.2, 21.6, 21.7, 23.1, **23.7**, 24.2, 24.2, 25.5, 26.6	*(Avg: 26.7)*	26.8, 28.9, 29.0, 29.7, **29.7**, 30.2, 32.5, 33.7, 33.7

- Overall Median (M) = **26.7 cm** (average of 9th and 10th).
- Q_1**:** The median of the first 9 observations. The middle of 9 is the 5th value $\rightarrow$ **23.7 cm**.
- Q_3**:** The median of the upper 9 observations. The middle of 9 is the 5th value $\rightarrow$ **29.7 cm**.

Example 2.6: Finding Quartiles in R

R Code

```
summary(torrey_needles)
```

Output:

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  21.20   23.82   26.70   27.28   29.53   33.70 
```

Software Note: Quartile Algorithms in R

Why does R give a slightly different answer ($Q_1 = 8.6$) than our hand calculation ($Q_1 = 8.5$)?

- *Reason:* There are actually 9 different mathematical algorithms for interpolating quartiles when the splits aren't perfectly even!
- Our hand-calculation rule is simply the easiest way to do it manually.
- R uses a more advanced interpolation method by default.
- **Best Practice:** When analyzing real biological data in R, always trust the output R provides!

The Interquartile Range (IQR)

The distance between the first and third quartiles is called the **Interquartile Range (IQR)**.

$$IQR = Q_3 - Q_1$$

- The IQR is the exact span of the **middle 50%** of your data.
- Because the IQR completely ignores the bottom 25% and top 25% of the data, it is a highly **resistant** measure of spread.
- A giant outlier in the tail will not affect the width of the middle 50%.

Calculating the IQR in R

R has a built-in function to instantly calculate the exact Interquartile Range.

R Code

```
IQR(pine_needles)
```

Output: `[1] 1.95`

Note: R uses its default quartile algorithm to compute this, so $10.55 - 8.60 = 1.95$.

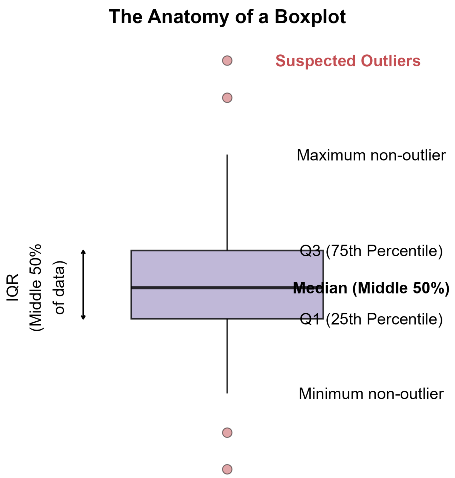
The Five-Number Summary

By combining our measures of center, our measures of spread, and the absolute extremes of our data, we get a highly efficient snapshot of the entire distribution called the **five-number summary**:

$$\text{Minimum} \quad Q_1 \quad \text{Median} \quad Q_3 \quad \text{Maximum}$$

These five numbers offer a reasonably complete mathematical description of a dataset's center and spread.

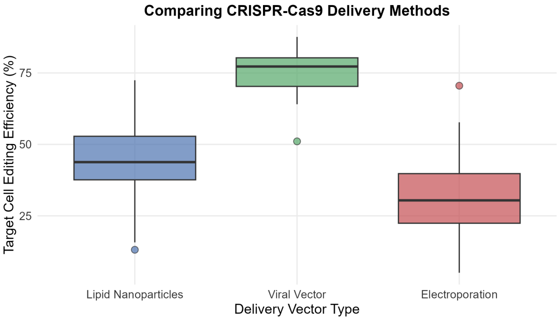
Boxplots: Visualizing the Five-Number Summary



How to Read a Boxplot:

- **Box:** Spans from Q_1 to Q_3. Its width is the exact **IQR** (middle 50%).
- **Thick Line:** The **Median**.
- **Whiskers:** Extend to the smallest and largest observations *that are not considered outliers*.
- **Dots:** Suspected **outliers** floating beyond the whiskers.

Side-by-Side Comparisons



Why are boxplots so popular? They allow us to compare dozens of groups cleanly in a single graphic.

- **Center:** Viral vectors yield the highest median efficiency.
- **Spread:** Electroporation is highly unstable (tall IQR box).
- **Outliers:** Lipid nanoparticles are consistent but have two high-performing outliers.

Identifying Outliers: The $1.5 \times IQR$ Rule

In Chapter 1, we said an outlier is an observation that falls "well outside" the overall pattern. But how does **R** automatically know which dots to plot outside the whiskers? Statisticians use an objective mathematical formula called the $1.5 \times IQR$ **Rule**.

Boundary	**Formula**
Upper Fence	$Q_3 + (1.5 \times IQR)$
Lower Fence	$Q_1 - (1.5 \times IQR)$

Any data point larger than the Upper Fence or smaller than the Lower Fence is plotted as an individual dot.

Exam Warning: Do Not Automatically Delete!

Warning on Outlier Deletion

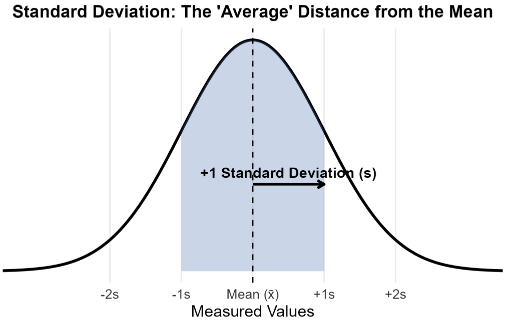
The $1.5 \times IQR$ rule tells you *that* a data point is mathematically unusual, but it does **NOT** give you permission to delete it!

- As we learned in Chapter 1, you must investigate the biology or lab methodology before removing any outlier.
- Many important biological discoveries (like super-responders to a drug) begin as dots outside the whiskers!

Measuring Spread: The Standard Deviation (s)

If we use the Median to measure center, we use the IQR to measure spread. But if we use the **Mean ($\bar{x}$)** to measure center, we need a different measure of spread: the **Standard Deviation (s)**.

- The standard deviation measures spread by looking at how far each observation falls from the mean.
- Conceptually, think of it as the **"average distance"** of all data points from the center.



The Math Behind Standard Deviation

To calculate how far the data is spread out, we find the deviation of each point: $(x_i - \bar{x})$.

- *Problem:* If we just added these up, negative distances would cancel out positive distances, summing to exactly zero.
- *Solution:* We **square** the distances to make them positive, average them, and then take the **square root** to return to original biological units.

Variance (s^2):

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

Standard Deviation (s):

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

Degrees of Freedom: Why $(n-1)$?

Why do we divide by $(n-1)$ instead of n?

- This is a concept called **degrees of freedom**.
- Because we already used our sample data to estimate the mean $(\bar{x})$ in the formula, we "spent" one piece of independent information.
- Dividing by $(n-1)$ corrects a mathematical bias, ensuring our sample standard deviation perfectly estimates the true underlying population spread.

Example 2.7: Calculating SD (Metabolic Rates)

A person's metabolic rate is the rate at which the body consumes energy. Here are the rates of 7 men in a dieting study (kilocalories/24-hour).

Raw Data ($n = 7$): 1792, 1666, 1362, 1614, 1460, 1867, 1439

Step 1: Find the Mean ($\bar{x}$)

$$\bar{x} = \frac{1792 + 1666 + 1362 + 1614 + 1460 + 1867 + 1439}{7} = \frac{11{,}200}{7} = 1600 \text{ Cal}$$

Example 2.7: Find Deviations and Square Them

Subtract the mean (1600) from every observation, and square the result.

Observation (x_i)	**Deviation** $(x_i - \bar{x})$	**Squared Deviation** $(x_i - \bar{x})^2$
1792	$1792 - 1600 = 192$	$192^2 = 36{,}864$
1666	$1666 - 1600 = 66$	$66^2 = 4{,}356$
1362	$1362 - 1600 = -238$	$(-238)^2 = 56{,}644$
1614	$1614 - 1600 = 14$	$14^2 = 196$
1460	$1460 - 1600 = -140$	$(-140)^2 = 19{,}600$
1867	$1867 - 1600 = 267$	$267^2 = 71{,}289$
1439	$1439 - 1600 = -161$	$(-161)^2 = 25{,}921$
	Sum = 0	**Sum of Squares = 214,870**

Notice the un-squared deviations sum to exactly zero!

Example 2.7: Calculate Variance and SD

Step 3: Calculate Variance (s^2)
Divide the sum of squared deviations by $n - 1$ (which is $7 - 1 = 6$):

$$s^2 = \frac{214{,}870}{6} = 35{,}811.67 \text{ Cal}^2$$

Step 4: Calculate Standard Deviation (s)
Take the square root of the variance to return to the original units:

$$s = \sqrt{35{,}811.67} = 189.24 \text{ Cal}$$

Conclusion: The men burn an average of **1600 Cal** per day, and their rates typically vary from that mean by about **189.24 Cal**.

Example 2.7: Calculating Variance and SD in R

R Code

```
# Create a vector of the 7 metabolic rates
metabolic_rates <- c(1792, 1666, 1362, 1614, 1460, 1867, 1439)

# Calculate Variance (s^2)
var(metabolic_rates)

# Calculate Standard Deviation (s)
sd(metabolic_rates)
```

Output:

```
[1] 35811.67
[1] 189.2397
```

Properties of the Standard Deviation

- **Measures spread around the mean:** It should only be used when $\bar{x}$ is chosen as the measure of center.
- **$s \geq 0$:** $s = 0$ only when there is zero spread (all observations are identical). Otherwise, $s > 0$.
- **Same units as original data:** If you measure tumor diameter in millimeters (mm), s is in mm. (Variance is in mm^2, which is biologically unhelpful).
- **Non-resistant!** Because the formula uses the mean and squares the distances, a single extreme outlier will inflate s drastically.

Choosing the Right Summary

You now have two complete sets of tools to summarize a quantitative distribution. How do you choose? It depends entirely on **Shape** and **Outliers**!

When to Use	**Mean & Standard Dev.**	**Median & IQR (5-Number)**
Shape	Reasonably Symmetric	Skewed (Left or Right)
Outliers	No extreme outliers	Contains suspected outliers
Why?	**Non-resistant.** Powerful for normal biology, but easily ruined by extreme values.	**Resistant.** Provides a stable snapshot even when outliers are present.

Summary of R Commands (Chapter 2)

Here are the key functions introduced in this chapter for calculating numerical summaries of a dataset (vector) `x`:

R Command	What it Calculates
`c(1, 2, 3)`	Combines individual data values into a single vector.
`mean(x)`	Calculates the sample mean ($\bar{x}$)
`median(x)`	Calculates the exact median (M)
`summary(x)`	Outputs the five number summary (Minimum, Q_1, Median, Q_3, Maximum) plus the mean
`IQR(x)`	Calculates the Interquartile Range ($Q_3 - Q_1$)
`var(x)`	Calculates the sample variance (s^2)
`sd(x)`	Calculates the sample standard deviation (s)

Chapter 2 Summary: Descriptive Statistics Checklist

When asked to summarize a quantitative dataset, always follow this workflow:

1. **Plot the data first:** Use a histogram or dotplot.
2. **Choose your tools:**
   - Symmetric $\rightarrow$ use **Mean** ($\bar{x}$) and **Std Dev** (s).
   - Skewed / Outliers $\rightarrow$ use **Median** (M) and **IQR**.
3. **Calculate resistance:** Remember the mean is pulled toward the long tail.
4. **Visualize comparisons:** Use side-by-side **boxplots**.
5. **Check for outliers mathematically:** Use the $1.5 \times IQR$ **rule** to flag unusual points, but investigate scientifically before excluding!

What's Next? Looking Ahead to Chapter 3

- In Chapters 1 and 2, we explored data that had only **one variable** recorded per individual (e.g., just body weight, or just viral load).
- But in biology, we often want to know if two different variables are connected:
    - *Does a higher dosage of a drug cause a faster drop in blood pressure?*
    - *Is a specific genetic mutation linked to a higher risk of heart disease?*
- In **Chapter 3: Scatterplots and Correlation**, we will learn how to visualize and mathematically measure the relationship between **two quantitative variables** measured on the same individual.